# Temporal anti-parity–time symmetry offers new way to steer energy through systems

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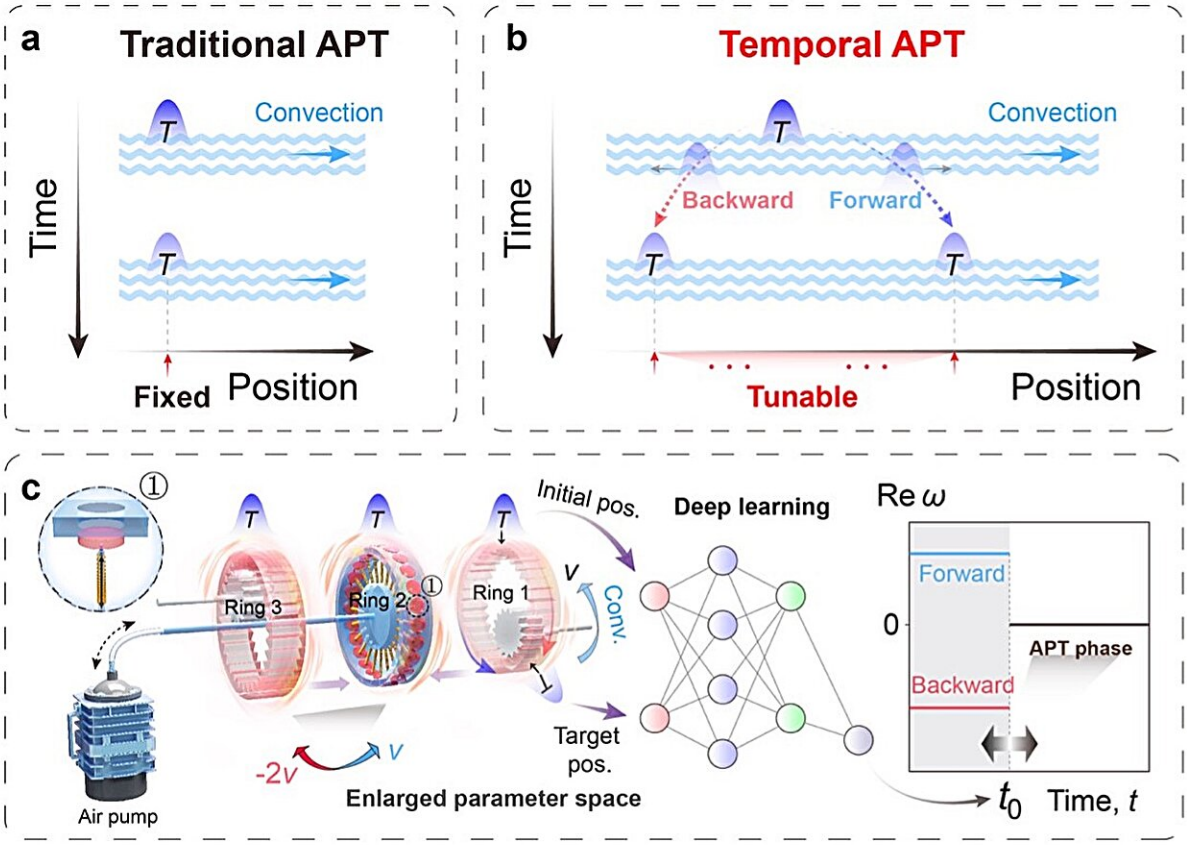


Engineering temporal anti-parity–time symmetry. a, Conventional anti-parity–time (APT) symmetry: convection is suppressed to achieve localized trapping, and the thermal wave packet gradually decays over time. b, Temporal anti-parity–time symmetry: under identical convective conditions, the temperature wave packet can be programmably transported forward (with convection) or backward (against convection), and trapped at arbitrary locations. c, Experimental platform and timing protocol: three concentric rotating rings

with synchronized switching of embedded medium and rotation speed; deep learning predicts the optimal switching time. Credit: Jin et al. *Nature Physics* (2025). DOI: 10.1038/s41567-025-03129-8.

The movement of waves, patterns that carry sound, light or heat, through materials has been widely studied by physicists, as it has implications for the development of numerous modern technologies. In several materials, the movement of waves depends on a physical property known as parity-time (PT) symmetry, which combines mirror-like spatial symmetry with a symmetry in a system's behavior when time runs forward and backwards.

Systems with PT symmetry can suddenly alter their behavior when they pass specific thresholds known as phase transitions, where they shift from balanced to unbalanced states. So far, systems exhibiting PT symmetry are mostly static, meaning that they exhibit fixed properties over time.

In *Nature Physics*, researchers at University of Shanghai for Science and Technology, Fudan University and National University of Singapore introduce a new concept called temporal anti-parity–time (APT) symmetry, which delineates more clearly both where and when a phase transition happens in a non-Hermitian system, a system that exchanges energy with its surroundings.

Their paper could open new possibilities for controlling how heat and energy move through materials over time.

"Our work was motivated by a clear gap between what non-Hermitian physics promises and what had been practically achieved in diffusive transport," Jiping Huang, senior author of the paper, told Phys.org.

"In wave systems, PT symmetry and exceptional points (EPs) have become powerful tools for controlling propagation. In diffusion—especially heat—recent progress enabled fascinating static effects such as localization and topological thermal states, but the control paradigm was still largely 'design it once and it stays that way.'"

Huang and his colleagues drew inspiration from optical tweezers, experimental tools that use highly focused laser beams to trap and move molecules, cells or other tiny objects. Rather than creating a static landscape, these tools allow researchers to catch, move and hold objects at will.

"We asked whether a similar capability could be achieved for energy—in our case, a temperature profile in a convective-diffusive system," said Huang.

"The primary objective thus became to make the timing of a non-Hermitian phase transition a programmable control knob, rather than treating the system as locked in a fixed symmetric or symmetry-broken phase.

"Concretely, we aimed to demonstrate that one can transport a temperature 'wave packet' forward with convection, transport it backward against convection and trap it at an arbitrary position—all by an on-demand, non-periodic temporal protocol."

## The introduction of temporal anti-parity–time symmetry

APT symmetry is a well-established rule that applies to non-Hermitian systems, specifically describing how they behave when space and time are reversed. APT symmetry entails that gain and loss in a system are

arranged oppositely to how they would be arranged in the presence of PT symmetry.

This balanced exchange structure can result in a so-called exceptional point (EP), a point at which a system transitions between two qualitatively different dynamical regimes. Huang and his colleagues applied this concept to the transport of heat through materials, to better predict and control the movement of heat over time.

"Instead of building a system that sits permanently in one regime, we actively switch the system in time so that the moment when it crosses the EP is precisely controlled. This makes the transport history itself programmable," explained Huang.

"A simple operational picture is a two-stage time script. During stage 1 (transport phase, before the switching time ), parameters are set so the temperature profile drifts either forward (with convection) or backward (against convection). During stage 2 (trapping phase, after ), we switch the system into an APT-symmetric configuration beyond the EP threshold, which locks and localizes the temperature modulation at the desired target position."

To demonstrate their concept in an experimental setting, the researchers used a three-ring thermal device. This device is comprised of two outer copper rings that rotate, pushing heat around in specific directions, as well as a middle ring that controls how easily heat jumps between the other two moving rings, which can be adapted over time. The middle ring acts as a time-tailored coupling layer with thermal properties that can be abruptly altered using a pneumatic actuator.

"At the switching time , we simultaneously change (i) ring rotation settings and (ii) the middle-ring embedded material, thereby changing the inter-ring heat exchange rate across the EP threshold," said Huang.

"Using infrared thermography, we showed that the temperature peak could be transported and then trapped at different prescribed locations (e.g., ~one-third-cycle or ~two-thirds-cycle away from the heating point). We also demonstrated the counterintuitive case where the temperature profile moves against the convection direction before being trapped."

The researchers wanted to determine the exact moment in which they should flip the direction of heat in their device to ensure that heat ended up at a desired location. Predicting this point in time, also known as the optimal , is difficult when running conventional analyses.

They thus employed a deep-learning model, using it to map the initial peak position and target position that would result in switching time . This allowed them to precisely control the movement of heat in their device.

## New routes for controlling the flow of heat

This recent study introduces a new approach to control the transport of energy in non-Hermitian systems with remarkable precision. In the future, their proposed strategy could be applied to the engineering of new technologies and devices in which heat moves following desired patterns.

"We extended non-Hermitian control in diffusion from a static phase viewpoint to a time-programmable viewpoint," said Huang. "The key shift is that we treat 'when the phase transition happens' as an engineerable degree of freedom."

Notably, Huang and his colleagues have realized a capability that is very difficult to demonstrate experimentally when employing conventional thermal management strategies. This is the on-demand bidirectional

transport and trapping of a specific temperature profile in a system in which heat would normally be carried a way and dissipate.

"In our platform, the temperature peak can move forward, move backward, or be immobilized at arbitrary locations, under the same overall geometry, by changing only the time protocol," said Huang.

"Finally, we provide a complete framework that integrates (i) non-Hermitian theory, (ii) finite-element simulation, (iii) deep-learning-assisted design of the switching time, and (iv) experimental validation, including robustness tests. We also demonstrated an application-oriented proof-of-concept: using trapped thermal dynamics to achieve an approximately 5 K enhancement in targeted cooling performance in a device scenario."

This research team's effort could soon pave the way for the intelligent routing of heat in different types of devices, as opposed to the passive heat diffusion observed in most existing technologies. This could enable the creation of microelectronics, in which heat flows precisely where it is needed, while target areas are cooled down.

"The idea of temporal control of EP crossings is a general blueprint that could be extended to other platforms—photonic, acoustic, or magnonic systems—where one can modulate material parameters or gain/loss in time," added Huang.

"In our next studies, we plan to extend our framework from one-dimensional (ring-like) transport to richer two-dimensional and three-dimensional spatiotemporal control, including steering around obstacles and programmable trajectories. We will also explore whether 'temporal non-Hermitian symmetry engineering' can be translated to other diffusive processes (mass diffusion, chemical transport) and, conversely, to wave systems where real-time modulation is feasible."

The researchers are now also working on additional machine-learning tools that could further improve their experimental approach. For instance, they would like to develop techniques that do not only output the optimal switching time, but can also co-optimize geometry, materials, and temporal protocols under practical constraints.

**More information:** Peng Jin et al, Temporal anti-parity–time symmetry in diffusive transport, *Nature Physics* (2025). DOI: 10.1038/s41567-025-03129-8.

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